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Mechanistic analysis of coal wettability enhancement by nanoparticle-assisted surfactant systems in aqueous/salt solution environment



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Highlights

- The coal wettability can be improved by salt solution combined with surfactants and nanoparticles.
- The potential of mean force, binding and dissociation energy barriers are analyzed from microscopic perspective.

- The relationship between the wettability of nanoparticles synergistic different types of surfactants is obtained.
- The interaction between metal cation, non-metal anion and inorganic salt is analyzed in depth.

Abstract

This paper explores patterns of coal wettability regulation through the synergistic action of nano-SiO₂ with cationic, anionic and nonionic surfactants under the actual coal seam environment of aqueous/salt solutions. Using molecular dynamics, models of the coal wall, aqueous solutions and salt solutions were constructed. The wetting characteristics of these systems were comprehensively investigated by examining the modification of different surfactants by nano-SiO₂, and incorporating common salt solutions (Na⁺, Ca²⁺, Mg²⁺, Cl⁻) typically present in coal seams. The results indicate that SiO₂@SDBS and SiO₂@DTAB enhance H₂O adsorption on the coal wall surface, and their water molecule diffusion coefficients are higher than that of SiO₂@NPEO. The analysis of radial distribution function and potential of mean force (PMF) reveals that the peak values of the radial distribution curves between SiO₂@SDBS, SiO₂@DTAB and H₂O are high, indicating stronger hydration and a more stable wettability effect. The dissociation energy barrier with SiO₂@SDBS is also the largest. SiO₂@DTAB exhibits the highest peak radial distribution value with Cl⁻, suggesting the strongest interactions. The g(r) of the radial distribution curves between SiO₂@surfactant and metal ions, as well as inorganic salts, all increase gradually. For the MgCl₂ system, the interaction of SDBS with Cl⁻ is the most significant. For the CaCl₂ system, NPEO has the strongest effect on Cl⁻. Compared with NaCl, MgCl₂ and CaCl₂ both have more significant effects on coal wettability when they work in synergy with different types of surfactants modified by nano-SiO₂. This study innovatively reveals the microscopic mechanisms by which the three surfactants synergistically regulate coal wettability with nano-SiO₂ in aqueous/salt solution environments, providing key theoretical support to clarify the influence of the coal seam environment on the performance of fracturing fluids, and enhancing the efficiency of gas dissolution and extraction.

Keywords

Aqueous solution; Salt solution; Wettability; Nanoparticle; Surfactant; Molecular dynamics

1. Introduction

Coal has played the role of a “ballast stone” in China’s energy structure. Efficient extraction and utilization of coal-associated gas can prevent mine gas accidents, reduce greenhouse gas emissions, improve energy supply structure, and promote the realization of dual-carbon goals [1,2]. However, most coal seams in China generally suffer from poor permeability, resulting in low efficiency of conventional borehole pre-drainage for gas. Therefore, high-efficiency permeability enhancement technology should be adopted [3, 4, 5]. Hydraulic fracturing is currently one of the effective measures for increasing permeability in coal seams and increasing coalbed methane production [6], its construction effect largely depends on the performance of fracturing fluids [7,8]. In recent years, the application of “nano-hydrocarbon” technology in hydraulic fracturing has attracted wide attention. Some scholars have added nanoparticles to fracturing fluid and found that the special properties of nanoparticles can optimize the performance of fracturing fluid [9]. Notably, coal seam wettability is crucial for fracturing fluid flowback efficiency and reservoir damage prevention. Efficient flowback of fracturing fluids can significantly reduce reservoir retention damage. By regulating wettability, liquid retention can be reduced, flowback efficiency can be improved, plugging and damage to reservoir pores can be reduced, and desorption and migration efficiency of coalbed methane can be improved [10,11].

Nanoparticles have demonstrated remarkable capabilities for performance enhancement in a wide range of fields, owing to their distinctive physicochemical properties like high specific surface area and small size effect [12, 13, 14]. In pure water systems, nano-SiO₂ regulates wettability by interacting with the surface functional groups of coal (such as hydroxyl and carboxyl) through hydrogen bonding, or by altering the coal-water interface energy through physical adsorption [15,16]. Zhao et al [17] studied the enhancement effect of SiO₂-H₂O nanofluids on the wettability of pulverized coal, finding that when the nanofluid concentration is 1.5wt%, the wettability of pulverized coal increased by 44.61%. Zhang et al. [18] tested the wetting effect of water-based SiO₂ nanofluids on coal at different concentrations by contact angle, and found that increasing nanofluid concentration improved water absorption capacity and coal wettability. The wetting effect of nanofluids gradually deteriorated over time, but the stability of effect gradually improved. Zou et al. [19] studied the regulation effect of water-based SiO₂ nanofluids on the water absorption and surface wettability of raw coal, systematically investigated the influence of time and water content on wettability, and

revealed the deep mechanisms by which SiO₂ nanofluids enhanced the effectiveness of coal seam water injection by optimizing the coal wettability. This research hotspot promotes the ability of water-based SiO₂ nanofluids to improve the wettability of coal seams.

Surfactants, on the other hand, are extensively employed due to properties including their ability to reduce surface tension and alter wettability [[20], [21], [22]]. Jiang et al. [23] discovered that they function as tunneling agents between the coal and water, thereby enhancing the hydrophilicity of bituminous coal. Ji et al. [24] used zeta potential experiments to show that the addition of surfactants improved coal dust aggregation ability and promoted the dust settlement in a water-based surfactant solution. Nanoparticles tend to produce more pronounced effects when they act in synergy with surfactants, initiating a series of intricate interactions and inducing microstructural alterations within the solution system. Verma et al. [25] found that Fe₂O₃ nanoparticle foam stabilizer enhanced the rheology of foams by selecting different combinations of nanoparticles and surfactants. Zhang et al. [26] explored the impact of deionized water on the wettability of coal treated with different solutions based on contact angle experiments, and found that nanoparticles significantly improved the wettability. Zan et al. [27] compared the adsorption and seepage characteristics of nano-emulsions together with surfactants within the pores of coal rocks and revealed the advantages of this combination in coal seam water injection. Zhou et al. [28] investigated how the interaction between sodium dodecyl sulfate (SDS) and titanium dioxide (TiO₂) nanoparticles impacts the viscosity of aqueous fluids, and discovered a synergistic effect when SDS and TiO₂ are added simultaneously at a specific temperature, resulting in a much lower viscosity compared to each component separately. Xiao et al. [29] studied the interaction of coal, water and SDBS modified nano-SiO₂ with different concentrations and their synergistic effect on coal wettability using molecular simulation, finding that the higher the SDBS concentration, the better the wettability of the coal-water interface.

During the formation and evolution of coal seams, groundwater and minerals carry cations such as Na⁺, Ca²⁺, Mg²⁺ through dissolution and exchange, which accumulate in pores and fissures to form a unique coal seams salt solution environment. The ionic composition and physico-chemical properties of the salt solution regulate the penetration and spread of fracturing fluid from the perspective of wettability by altering the role of nanoparticles-surfactants at the coal-liquid interface [[30], [31], [32]]. Al-Anssari et al. [33] focused on the influence of nano-SiO₂ on the capacity of surfactants to decrease the oil-water interfacial tension. Additionally, they also studied the synergistic effects among the nanoparticle-surfactant, salt-nanoparticle, and surfactant-salt-nanoparticle systems on the reduction of interfacial tension and the critical micelle concentration of surfactants at different temperatures and salinities. Rasoul et al. [34] explored the synergistic interactions among nano-SiO₂, bio-surfactant, and salinity on the oil recovery rate during oil displacement, and found a positive synergistic effect between bio-surfactant and

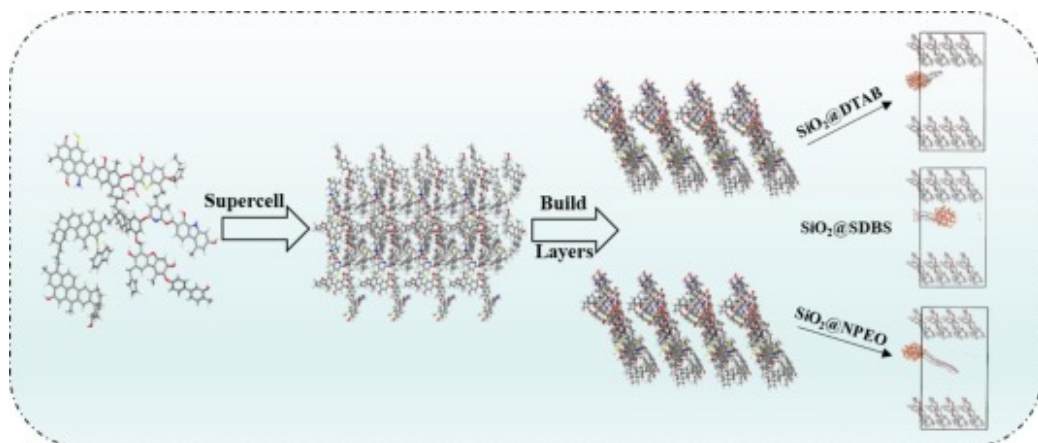
nano-SiO₂, enhancing the oil recovery rate by altering the wettability to water-wet conditions and reducing the interfacial tension. Devakumar et al. [35] studied the synergistic effect of low mineralization surfactant nanofluids on interfacial tension, wettability alteration and surfactant adsorption on quartz surfaces within oil–water systems, and determined that the wettability change was more significant in quartz substrates in sour oil environments, while surfactant adsorption decreased with increasing silica nanoparticle concentration.

In the actual coal seam environment, the interaction between the fracturing fluid and the coal is affected by the salt ions in the coal seam water (such as Na⁺, Ca²⁺, Mg²⁺, and Cl[−], etc.) [36]. Although existing studies have analyzed the synergistic effects of nano-SiO₂ and surfactants on regulating the coal wettability, but there is a lack of systematic research on the synergistic effects of different types of surfactants (cationic, anionic, nonionic, etc.) and nano-SiO₂. Meanwhile, the salt ions in the actual salt solution environment of the coal seam will affect the wetting characteristics. Therefore, this study systematically explores the synergistic effect of nano-SiO₂ with cationic, anionic, and non-ionic surfactants on coal wettability, focusing on the aqueous/salt solution environments. Using the Materials Studio (MS) software, it builds models for the coal wall, synergistic nanoparticle surfactant aqueous solutions, and salt solutions, and simulates interactions between the salt ions, nanoparticles, and different types of surfactants at the coal-water interface from the molecular level. The results are expected to provide theoretical support to further elucidate the influence of different coal seam environments on the performance of fracturing fluid and improve the efficiency of gas extraction.

2. Model construction and simulation methods

2.1. Coal wall models

The Wiser coal model has a wide distribution range of aromatic rings, which can fully consider the morphology of C, O, H, S and N atoms and the size of the aromatic nuclei in coal, more fully reflecting the diversity of its molecular structures [37]. The Visualizer module was used to build the Wiser coal model, and its molecular formula is C₁₉₁H₁₆₉N₃O₂₁S₃. The energy of the molecule was minimized through the Geometry Optimization and Anneal tasks within the Forcite module. Periodic boundary conditions were added to the molecules using Amorphous Cell, the density was 1.2g/cm³ and the force field was selected as COMPASS II. The cell was expanded to build a 4×2×2 supercell, and a 20 Å vacuum layer was added by Build Layers. This mode was used to analyze the properties of the coal surface and the pore characteristics of the coal body. The cell parameters of the resulting coal wall model are 63.85×31.92×87.49 Å³, as depicted in Fig. 1.



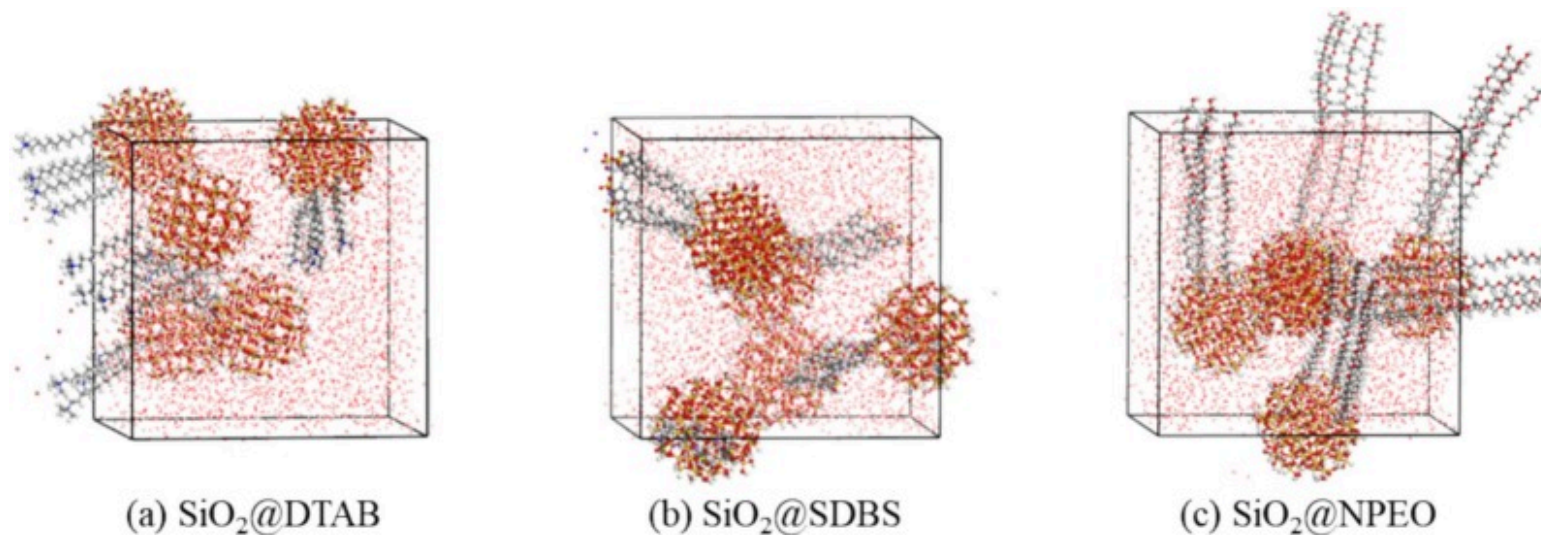
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Fig. 1. Construction of the coal wall models.

2.2. Nanoparticle synergistic surfactant aqueous solution models

The “Build Nanostructure” task was utilized to build nano-SiO₂ with a diameter of 1 nm for modification by three surfactants, used to investigate the microscopic mechanism of interaction between nanoparticle surfactant and water in an aqueous solution: Dodecyl Trimethyl Ammonium Bromides (DTAB) [38], Sodium Dodecylbenzene sulfonate (SDBS) [39] and Nonylphenol Ethoxylate (NPEO) [40], as representatives of cationic surfactants, anionic surfactants and nonionic surfactants, respectively. The Amorphous Cell module was used to build an aqueous box of SiO₂@DTAB, SiO₂@SDBS and SiO₂@NPEO containing 5 SiO₂@surfactant and 3500 water molecules. For this construction process, the COMPASS II force field was applied throughout, and the electrostatic and van der Waals potentials were computed by Ewald and Atom based methods, respectively, both with the cutoff radius set at 12.5 Å. The resulting aqueous solution models are shown in Fig. 2, and the respective cell sizes are 56.0×56.0×55.0 Å³, 55.7×55.7×56.0 Å³, and 56.5×56.5×56.0 Å³, respectively.



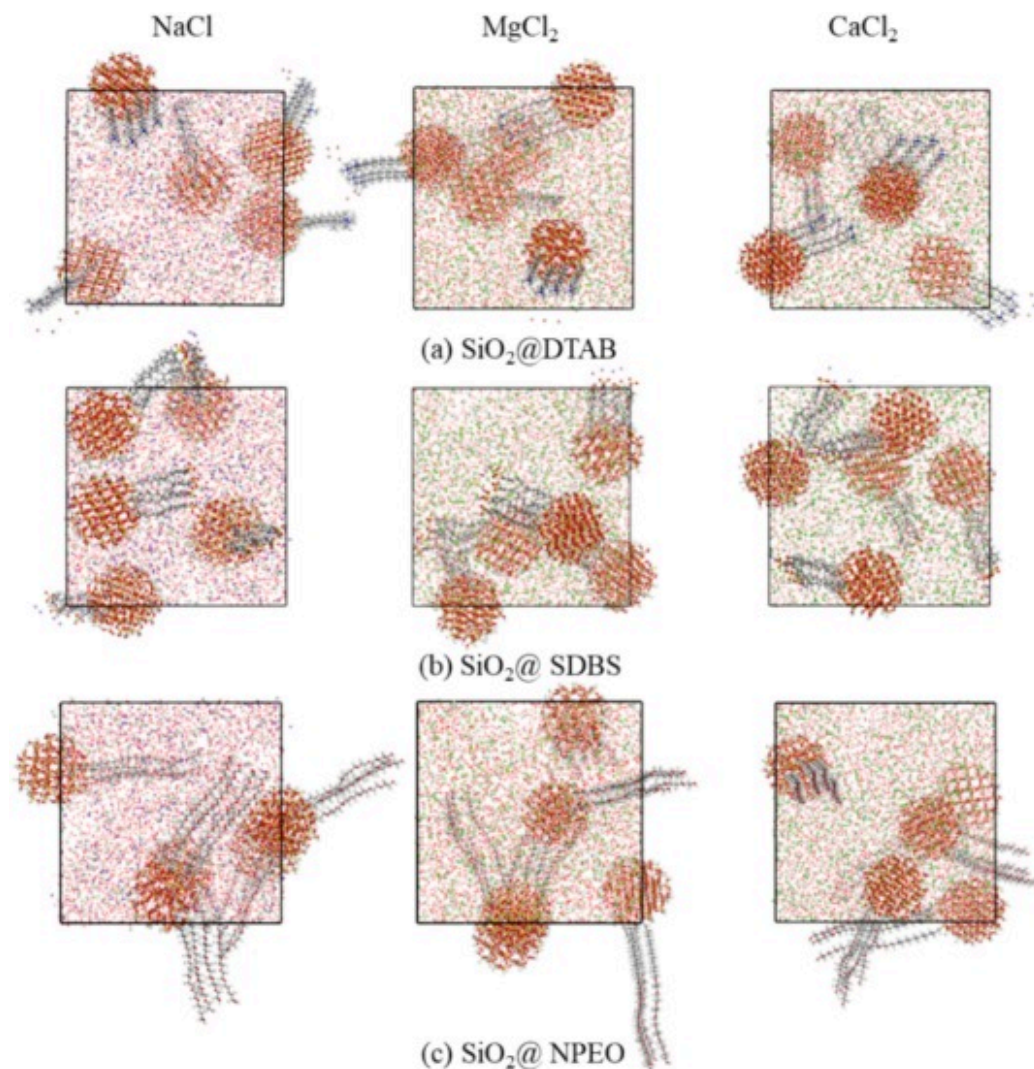
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Fig. 2. Aqueous solution models for SiO₂@surfactant.

2.3. Nanoparticle synergistic surfactant salt solution models

Common metal ions generally present in coal seams include Na⁺, Mg²⁺ and Ca²⁺ [36]. Inorganic salt solution models for nanoparticle synergistic surfactant were built to investigate the effect of salt solutions (NaCl, MgCl₂, and CaCl₂) in the presence of the three metal ions on the coal wettability with nanoparticle synergistic surfactants. Three salt solution models were built for each surfactant-modified nanoparticle, as shown in Fig. 3. Table 1 shows the specific information for each salt solution model.



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Fig. 3. Salt solution models for SiO₂@surfactant.

Table 1. Specific information for each salt solution simulation system.

	SiO₂@surfactant	Na⁺	Mg²⁺	Ca²⁺	Cl⁻	H₂O	size/Å³
SiO ₂ @DTAB@ salt solution	5	500	0	0	500	3500	60.7×60.7×60.0
	5	0	500	0	1000	3500	63.2×63.2×63.0
	5	0	0	500	1000	3500	64.3×64.3×64.0
SiO ₂ @SDBS@ salt solution	5	500	0	0	500	3500	60.8×60.8×60.0
	5	0	500	0	1000	3500	63.3×63.3×63.0
	5	0	0	500	1000	3500	64.4×64.4×64.0
SiO ₂ @NPEO@ salt solution	5	500	0	0	500	3500	61.0×61.0×61.0
	5	0	500	0	1000	3500	63.5×63.5×64.0
	5	0	0	500	1000	3500	64.5×64.5×65.0

2.4. Simulation methods

Adsorption studies were carried out using the Adsorption isotherm task within the Sorption module of MS, which were primarily executed through the Metropolis algorithm for the Monte Carlo method with 1×10^4 equilibration steps and 1×10^6 production steps. The electrostatic and van der Waals interactions were calculated using the Ewald and Atom Based methods, respectively, with a cutoff distance of 12.5 Å and calculation accuracy of Medium. The adsorption temperature was set to 298K, and the adsorption pressure ranged from 10kPa to 1000kPa.

Stable saturated adsorption configurations of the water molecules were obtained via the Locate task. Subsequently, Dynamics within the Forcite module was selected to perform kinetic calculations of H₂O in the coal wall model, as well as in the surfactant aqueous/salt solution models. The temperature and pressure control methods were Nosé-Hoover and Berendsen, respectively, and NVT was selected for the ensemble. The initial velocity is randomly assigned to reproduce the true motion of the particles and promote thermal equilibrium more quickly. The temperature was set at 298K. The COMPASS II force field was used in the calculation process, and the Ewald and Atom Based methods were adopted for electrostatic interaction and van der Waals

interaction, respectively, with 1×10^5 simulated steps, 100ps of simulation time, and a time step size of 1 fs. The data acquisition was performed every 1000 steps, and the resulting trajectory files were analyzed.

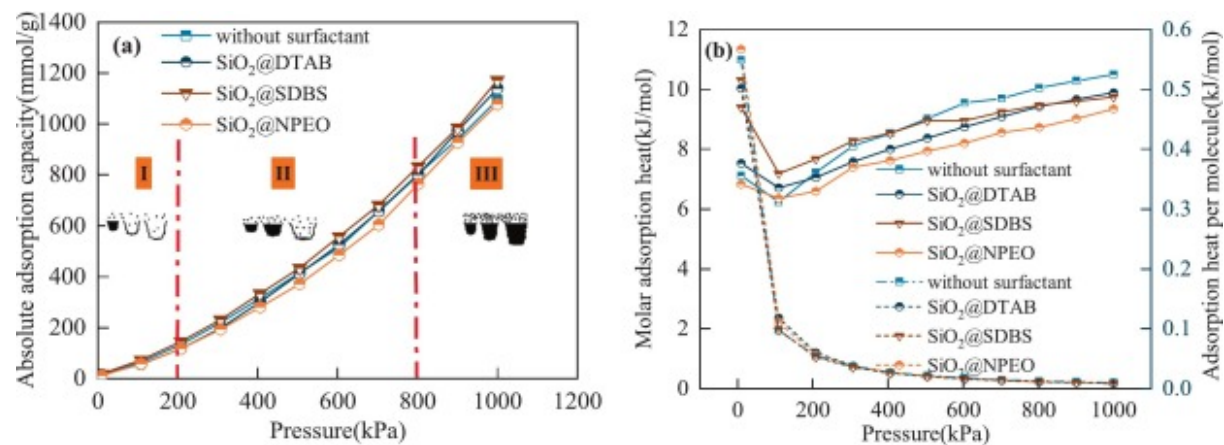
The electrostatic potential distribution on the surfaces of the coal and surfactant molecules were mainly investigated using the DMol3 module to find the self-consistent solution of the density functional theory (DFT) equations. The wave function and electron density were obtained, and used to derive various properties of the microscopic system. The Generalized Gradient Approximation (GGA) and BLYP functional were used for the wave function. The respective energy, interatomic force, and atomic displacement convergence accuracies in the geometric optimization process were 2×10^{-5} Ha, 0.004 Ha/Å and 0.005 Å, and the self-consistent convergence accuracy of electrons was 1.0×10^{-5} .

3. Results and discussion

3.1. The effect of synergistic nanoparticles surfactants on H₂O adsorption/diffusion on the coal surface

3.1.1. H₂O adsorption

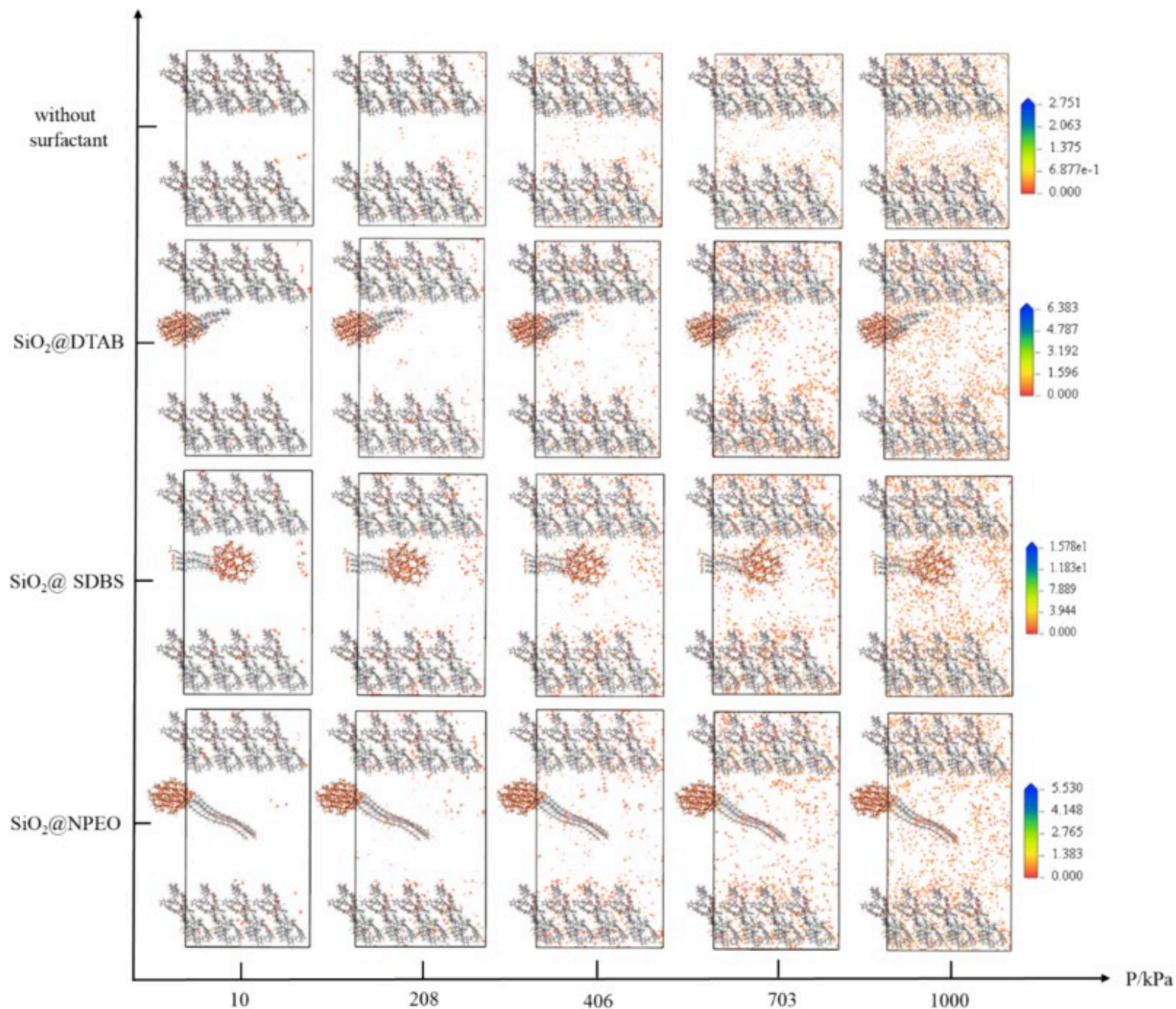
Fig. 4 shows H₂O adsorption on the coal wall surface at different pressures, and the magnitude of adsorption heat. With increasing pressure, H₂O adsorption is enhanced, and the relationship follows the following order: SiO₂@SDBS > SiO₂@DTAB > without surfactant > SiO₂@NPEO. Thus, SiO₂@SDBS and SiO₂@DTAB promote H₂O adsorption, while SiO₂@NPEO does not exhibit this effect. This finding accords with the adsorption density of H₂O on the coal wall surface at different pressures as presented in Fig. 5.



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Fig. 4. Adsorption capacity and adsorption heat of H₂O molecules on the coal wall surface at different pressures.



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Fig. 5. Adsorption density of H₂O molecules on the coal wall surface at different pressures.

The influence of oxygen-containing functional groups on H₂O adsorption mainly includes primary and secondary adsorption sites. The former mainly occur on solid surfaces where strong hydrogen bonding form between water molecules and hydrophilic pores. In secondary adsorption, in contrast, water molecules are usually adsorbed at sites with weaker binding energy. For the coal wall surface, H₂O adsorption is mainly affected by the coal matrix expansion effect [41], pore surface hydration [42], ink bottle pore effect [43] and interlayer cation hydration [44]. As shown in Fig. 4, Fig. 2, H₂O adsorption on the coal wall surface can be mainly categorized into three stages: Stage I (single-layer adsorption), Stage II (multi-layer adsorption) and Stage III (capillary condensation). Specifically, in Stage I, the majority of H₂O molecules are adsorbed in monolayer form through their strong hydrogen bonding with either the pore surfaces or oxygen-containing functional groups, which are the predominant primary adsorption sites. In Stage II, characterized predominantly by multilayer adsorption, as the distance between the pore surfaces (excluding the monomolecular layer) and water molecules increases, the interaction between them weakens. Concurrently, some water clusters begin to form within the hydrophobic pore space, occupying the pore volume which is initially accessible for water storage and movement. In Stage III, which is chiefly regulated by capillary condensation, H₂O get adsorbed at secondary adsorption sites. Due to strong hydrogen bonding, water clusters with highly ordered structures occupy an even larger part of the pore space.

The stronger the attraction between the coal and water, the smaller the wetting angle becomes, indicating enhanced wettability. When the adsorption capacity is weak, fine dust with a high degree of metamorphism is not wetted quickly and effectively. During hydraulic fracturing, this situation not only affects the infiltration of the fracturing fluid into the coal, but also makes it difficult for the fracturing fluid to fully penetrate the pores and fissures within it, thus weakening the fracturing effect. Given that nanoparticles effectively adsorb and coagulate dust particles, the synergistic use of nanoparticles with surfactants can facilitates H₂O adsorption onto the coal wall surface, thereby enhancing its wettability. This synergistic effect enables the fracturing fluid to penetrate the coal more uniformly, and also promotes the expansion of fissures within the coal body, creating a more unobstructed pathway for gas desorption and seepage, which both enhances gas extraction, and reduces the risk of gas outburst.

Adsorption heat serves as a crucial parameter for distinguishing between physical and chemical adsorption. Generally, higher 42 kJ/mol is considered chemical adsorption, and less as physical adsorption [45]. Fig. 4(b) shows the magnitude of the H₂O

adsorption heat, indicating that it is physical adsorption. The adsorption heat increases gradually with increasing pressure, demonstrating that the interaction becomes more intense. After converting the adsorption heat into the standardized adsorption heat per water molecule, this value decreases as the pressure increases. This effect is because when the pressure rises, the number of water molecules adsorbed increases (seen Fig. 4(a)), the distance between water molecules decreases, and the repulsive force between them gradually increases. This force weakens the interaction between the water and the coal molecules, reducing the energy released by adsorbing an additional water molecule, and thereby lowering the standardized adsorption heat of each. From a thermodynamic perspective, in equilibrium, the system adapts to this change by reducing the adsorption heat of individual water molecules, preventing system instability caused by excessive heat release.

3.1.2. H₂O diffusion

The ability of water to diffuse through small pores and crevices in the coal can be characterized by mean square displacement (MSD). The diffusion coefficient can be determined by using MS to analyze the MSD curve of water molecules and performing a linear fit of this curve with respect to time (in the form of $y=ax+b$). The specific equation is as follows [46]:

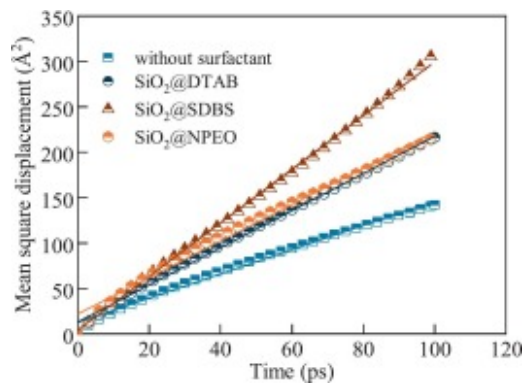
$$MSD(t) = \frac{1}{N} \sum_{i=1}^N \left(|r_i(t) - r_i(0)|^2 \right) \quad (1)$$

$$D = \lim_{t \rightarrow \infty} \frac{1}{6t} \left(\sum_{i=1}^N \left(|r_i(t) - r_i(0)|^2 \right) \right) = \frac{1}{6} K_{MSD} \quad (2)$$

where MSD is the mean square displacement, N is the number of particles of the same type, $r(t)$ and $r(0)$ are the position vectors of the molecules at time t and at the initial moment, respectively, D is the diffusion coefficient, t is the simulation time, and N_t is the number of steps in molecular dynamics, used for statistical averaging.

Fig. 6 presents the mean square displacement of H₂O on the coal wall surface, and Table 2 shows its linear fitting results. The diffusion coefficient of H₂O on the coal wall surface is the smallest when no surfactant is added, with a value of $2.23 \times 10^{-9} \text{ m}^2/\text{s}$. Nanoparticles synergized with different surfactants accelerate the diffusion. The SDBS anionic surfactant promotes diffusion most rapidly, followed by DTAB cationic surfactant, while NPEO nonionic surfactant results in the slowest diffusion, i.e. $\text{SiO}_2@\text{SDBS} > \text{SiO}_2@\text{DTAB} > \text{SiO}_2@\text{NPEO}$. This result is consistent with Zou et al. [47]. By preparing different surfactant-nanoparticle composite solutions, they found that the synergistic effect of nanofluids and surfactants improved the coal wettability in the contact angle test, infrared spectroscopy, and other characterization methods. Therefore, the application of nanoparticles in

combination with surfactants in hydraulic fracturing enhances the diffusion and penetration of fracturing fluid into the pores and fissures within the coal. This, in turn, enables the fracturing fluid to comprehensively and deeply infiltrate the coal body structure, completely filling its internal pores and fissures, and maximizing modification of its structure.



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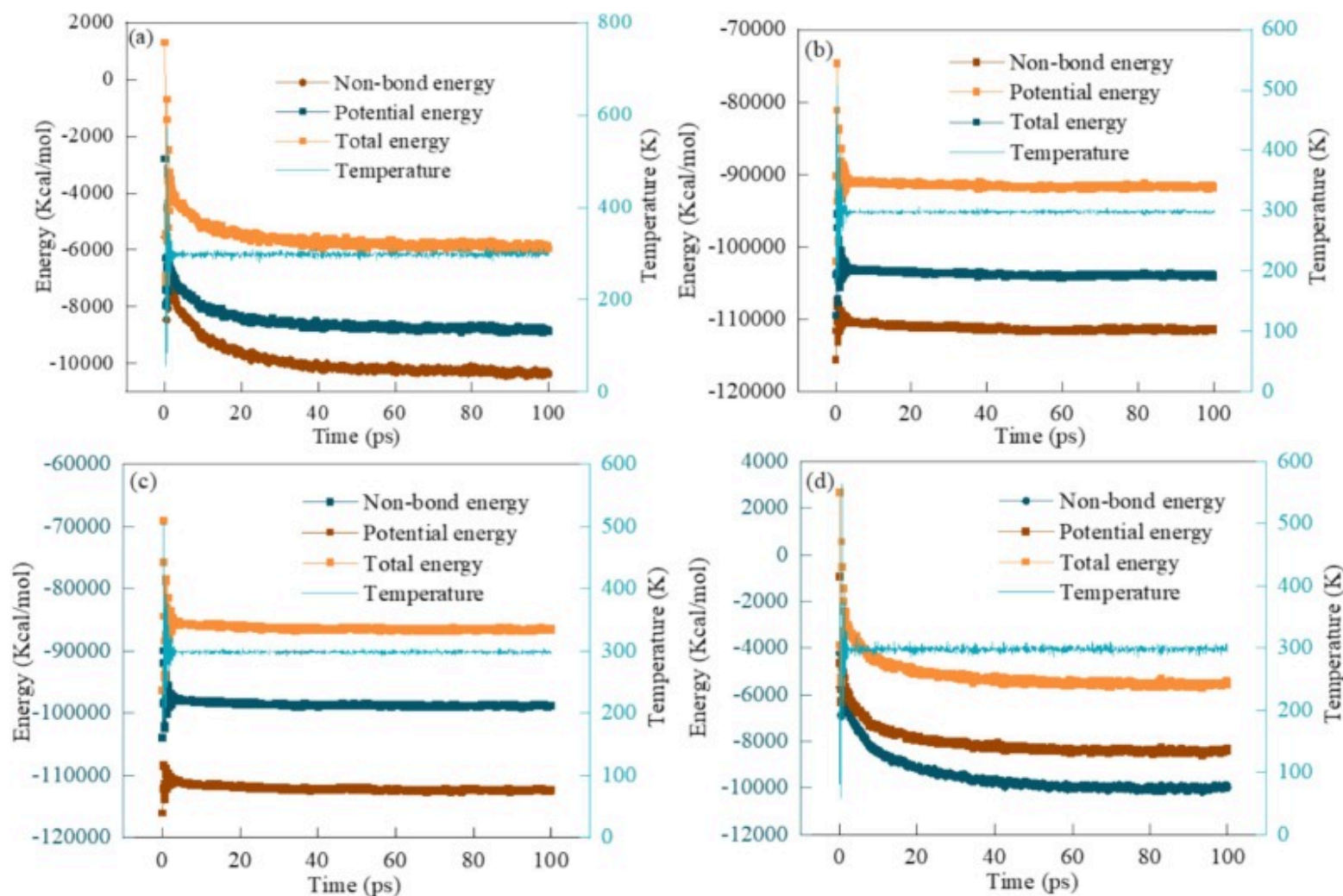
Fig. 6. Mean square displacement of H₂O on the coal wall surface.

Table 2. Fitted coefficient results of H₂O diffusion on coal wall surface.

	Linear fit formula	Coefficient of determination (R ²)	Diffusion coefficient (10 ⁻⁹ m ² /s)
Without surfactant	$y_1 = 1.34x_1 + 12.71$	0.994	2.23
SiO₂@DTAB	$y_3 = 2.05x_3 + 13.03$	0.998	3.42
SiO₂@SDBS	$y_2 = 2.95x_2 + 4.29$	0.999	4.92
SiO₂@NPEO	$y_4 = 2.01x_4 + 21.94$	0.991	3.35

After completing molecular dynamics simulations, it is necessary to determine whether the system has reached equilibrium. Generally, after the system reaches a stable state, both the temperature and energy parameters tend to be in equilibrium and do

not exhibit major changes. Fig. 7 shows the determination of the equilibrium parameters of the simulated system. Changes in the temperature, non-bond energy, potential energy, and total energy show that the dynamic process reaches equilibrium at 100ps.

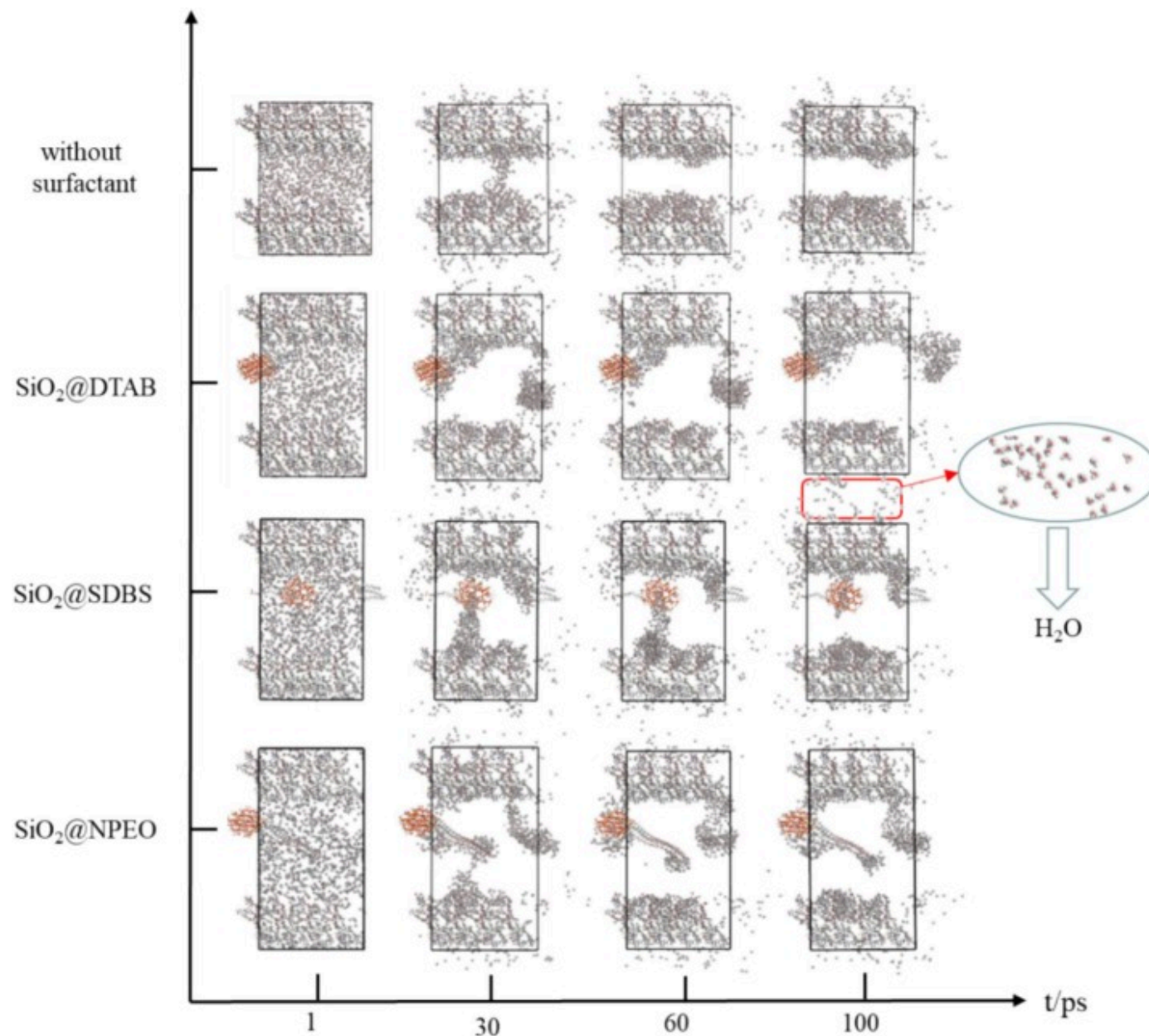


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Fig. 7. Determination of the equilibrium parameters of the simulated system ((a)-(d) are with no surfactant, SiO₂@DTAB, SiO₂@SDBS, and SiO₂@NPEO, respectively).

Fig. 8 displays the snapshot images of the water diffusion into the coal wall surface at various moments during the molecular dynamics process. At 1 ps, H₂O is saturated and adsorbed into the surface, as well as within the slit pores. As the kinetic process progresses, the H₂O molecules in the slit pores gradually migrate towards the coal wall surface. Additionally, synergistic nanoparticle surfactant drives the adsorbed H₂O to gradually migrate towards the coal wall surface. This increases the quantity of H₂O molecules adsorbed on the coal wall surface, thereby enhancing the wettability.

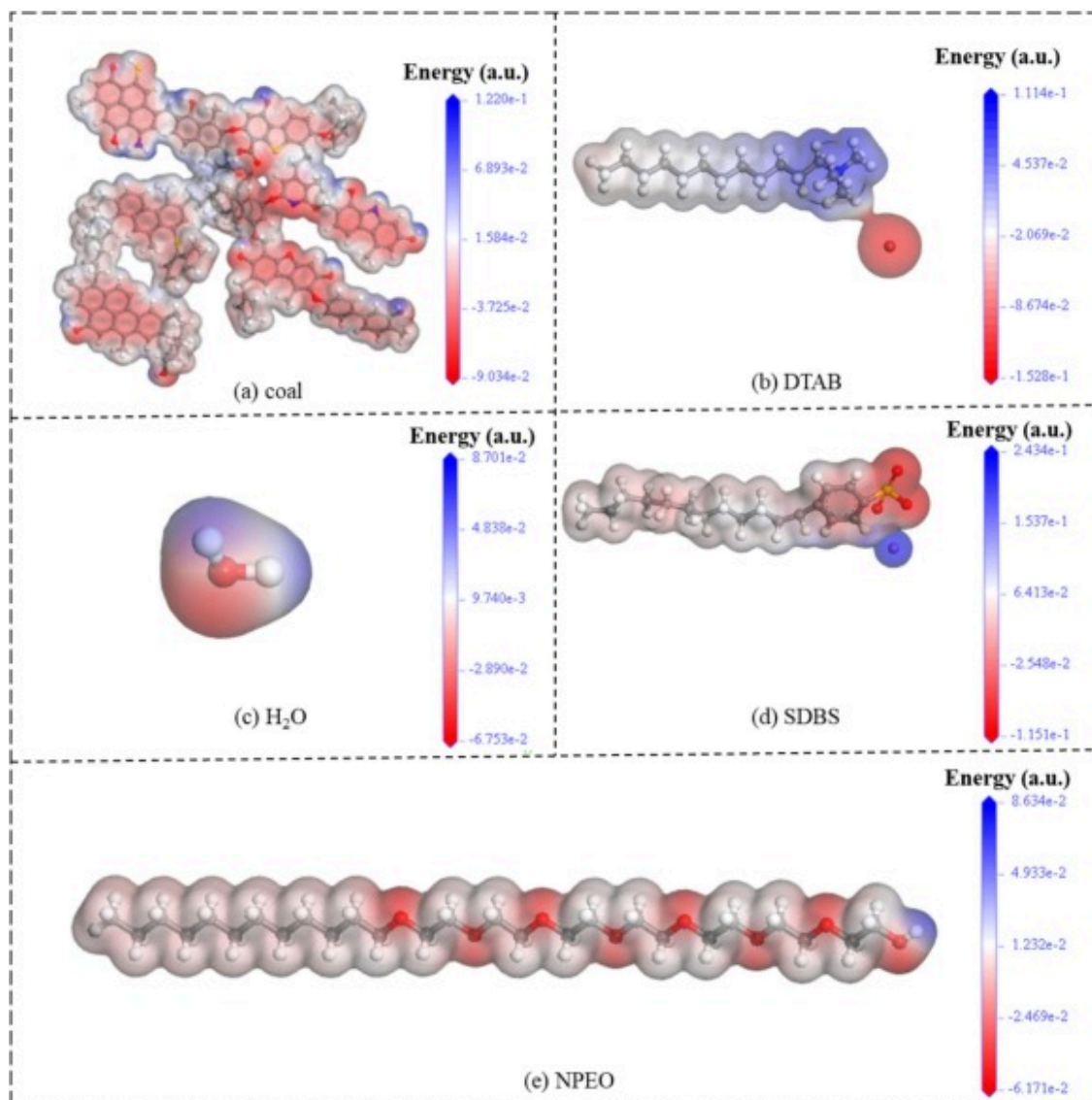


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Fig. 8. Snapshots of water molecule diffusion behavior on the coal wall surface.

The coal surface is anisotropic, irregular, and inhomogenous, which results in variations in the atomic microenvironment in different regions and locations. The interaction between diverse types of surfactants, water, and the coal arises from the electrostatic potential present on the surface of each molecule, as illustrated in [Fig. 9](#). Analyzing the electrostatic potential distributions of the surfactants, coal, and water can further our understanding of the primary factors contributing to the differences in coal wettability. [Table 3](#) shows the electrostatic potential on the surface of each molecule.



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Fig. 9. Surface electrostatic potential distribution.

Table 3. Electrostatic potential magnitude of each molecule.

	Negative potential/a.u.	Positive potential/a.u.
Coal	-0.090336	0.12201
H₂O	-0.067534	0.087015
DTAB	-0.1528	0.11142
SDBS	-0.11509	0.24335
NPEO	-0.061705	0.086337

Both the positive and negative potentials on the surface of H₂O molecules are smaller than those on the coal surface, indicating that H₂O is easily attracted to the coal molecules, thereby resulting in wetting. This is consistent with the electrostatic potentials of H₂O (positive is 0.0935 a.u.; negative is -0.0718 a.u.) and coal (positive is 0.1012 a.u.; negative is -0.0959 a.u.) calculated by Zhao et al. [48]. Under normal circumstances, the positive and negative electrostatic potentials of water molecules in aqueous solution attract each other and are closely connected through hydrogen bonding. The lowest electrostatic potential at the (Br⁺) cation in DTAB surfactant is -0.1528 a.u., while the maximum electrostatic potential is 0.11142 a.u. These values are respectively larger than the negative and positive potentials of water, thus DTAB has the potential to enhance wettability. Similarly, SDBS also enhances wettability; moreover, its effect is more pronounced than that of DTAB. In contrast, NPEO nonionic surfactant shows little wetting tendency, which is consistent with the adsorption conclusion in [Section 3.1.1](#).

3.2. Wettability analysis of aqueous solution on synergistic nanoparticle surfactants

3.2.1. RDF and coordination number distribution

The radial distribution function (RDF) describes the particle microstructure and reflects their aggregation characteristics, which mainly refers to the ratio of system localized density to the average bulk density [49]. It is calculated as follows:

$$g(r) = \frac{dN}{4\pi r^2 dr} \quad (3)$$

where $g(r)$ denotes the degree of aggregation, r is the distance from particle B to A, dN is the average number of B particles distributed in the range $r \sim r+dr$, and ρ represents the density of B.

In order to further characterize the interaction strength between SiO₂@surfactant and H₂O, the coordination number is calculated using the following formula:

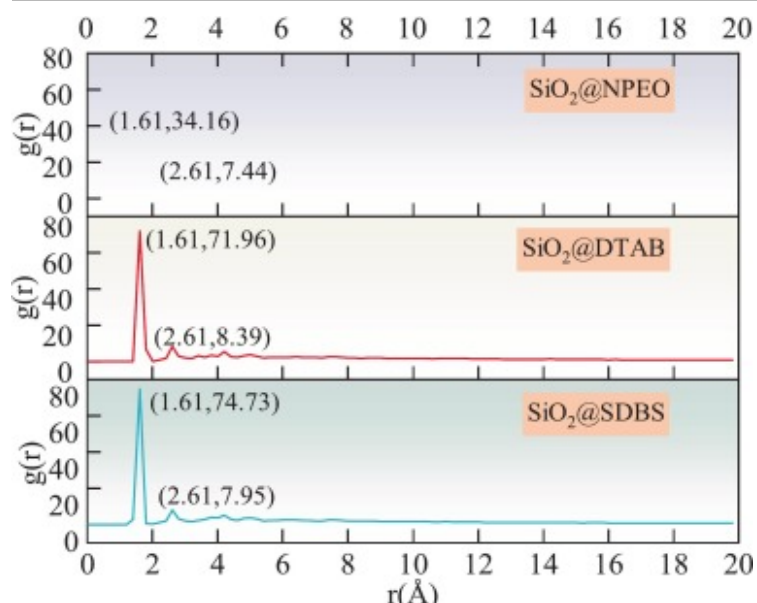
$$C_N(r) = \int \frac{g(r)N_B 4\pi r^2 dr}{V} \quad (4)$$

In the formula, N_B is the number of particles B surrounding particle A, and V is the volume of the periodic system model, 10^{-3} nm^3 .

Fig. 9 presents the radial distribution curves of SiO₂@surfactant with H₂O. The $g(r)$ curves have visible peaks located at 1.61 Å and 2.61 Å, which are the coordination shell radius of H₂O relative to the SiO₂@surfactant in the corresponding systems. The coordination shell radius of SiO₂@surfactant is similar, indicating similar distance characteristics of H₂O and the two surfactants. The peak value indicates the existence of a hydration layer surrounding the surfactant and nano-SiO₂. The width of the peak, meanwhile, reflects the degree of disorder in the coordination layer. Narrower peaks indicate relatively ordered coordination layers, and visa versa. The peak corresponding to SiO₂@SDBS is the narrowest, indicating a highly ordered hydration layer. Specifically, the peak for the cationic surfactant DTAB mainly results from the electrostatic interaction between its head group and the water. In the case of the anionic surfactant SDBS, its head group forms strong hydrogen bonding with the water, coupled with the electrostatic force, leading to the formation of a hydration layer. As for the nonionic surfactant NPEO, the formation of its hydration layer is primarily attributed to van der Waals interactions with the water. In addition, its peak value in the radial distribution curve with H₂O is lower than with SiO₂@SDBS and SiO₂@DTAB. The higher the peak value, the stronger the hydration between the two, and thus the more stable wetting effect.

The coordination number at the two peaks is calculated by observing the three $g(r)$ curves in Fig. 10, and the resulting coordination number is shown in Table 4. The coordination numbers of the first and second layer in SiO₂@DTAB and SiO₂@SDBS systems are larger than those of SiO₂@NPEO. The coordination number reflects the number of water molecules within a specific distance range. The higher the number, the denser the distribution of water molecules in the region, the stronger the interaction with the SiO₂@surfactant and water, the more stable the hydration, and the better the wettability. Through comprehensive analysis of $g(r)$ and coordination numbers, the ionic surfactants (SDBS and DTAB) form highly ordered and dense hydration layers

through strong electrostatic/hydrogen bonding, which significantly improves wettability. The nonionic surfactant (NPEO) has poor wettability and an unstable hydration layer due to weak interaction.



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Fig. 10. Radial distribution curves of SiO₂@surfactant with H₂O.

Table 4. Coordination number calculation results of H₂O with respect to SiO₂@surfactant.

	First coordination number	Second coordination number
SiO ₂ @DTAB	17.0	5.9
SiO ₂ @SDBS	16.0	5.0
SiO ₂ @NPEO	8.0	3.4

3.2.2. PMF

The Potential of Mean Force (PMF) characterizes how strongly two molecules interact with each other. The binding and dissociation energy barriers between molecules can be determined by the PMF. It is calculated based on the radial distribution function, as follows [50]:

$$\frac{W(r)}{k_B T} = \ln g(r) \quad (5)$$

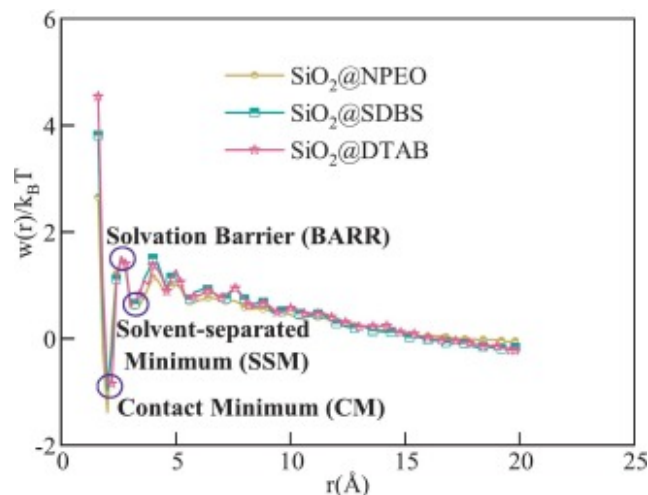
where $W(r)$ is the PMF; k_B is the Boltzmann constant, $1.3806505 \times 10^{-23}$ J/K; T is temperature, K; and $g(r)$ is a radial distribution function.

Fig. 11 depicts the PMF curve between SiO_2 @surfactants and H_2O . According to the literature [51], the PMF curve features three characteristic points, which are labeled in Fig. 11: the Contact Minimum (CM), Solvent-separated Minimum (SSM), and Solvation Barrier (BARR). The binding energy barriers between SiO_2 @surfactants and H_2O are determined by BARR and SSM, while the dissociation energy barriers depend on BARR and CM, as presented in Eqs. (5), (6).

$$\Delta E_1 = E_{BARR} - E_{SSM} \quad (6)$$

$$\Delta E_2 = E_{BARR} - E_{CM} \quad (7)$$

where ΔE_1 and ΔE_2 denote the binding energy barrier and dissociation energy barrier, respectively. A larger dissociation energy barrier indicates a stronger interaction between two entities, making its separation more difficult. Conversely, a smaller dissociation energy barrier implies a weaker interaction, requiring less energy for separation.



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Fig. 11. PMF curves between SiO₂@surfactant and H₂O.

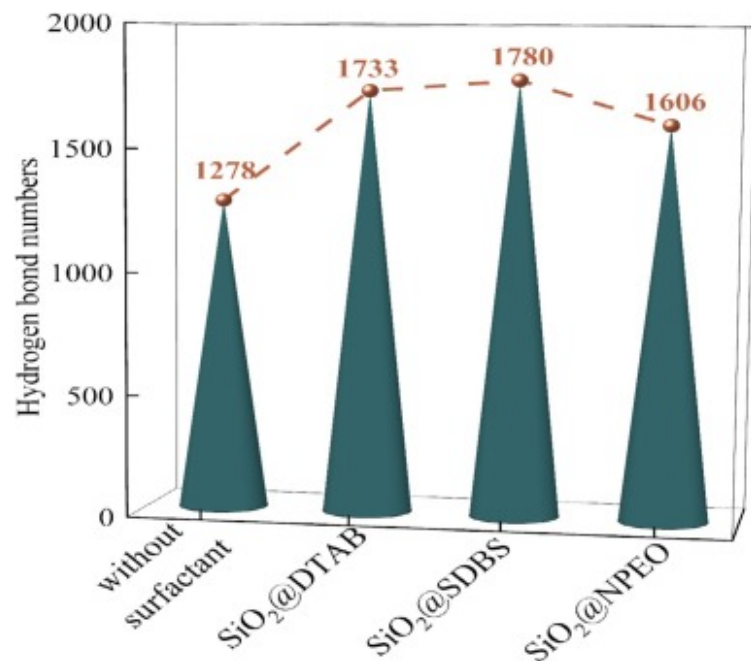
Table 5 shows the binding energy barriers and dissociation energy barriers between SiO₂@surfactants and H₂O. Due to the hydrogen bonding between SDBS anionic surfactant and water, the dissociation energy barrier (ΔE_2) of SiO₂@SDBS is the largest, at 2.88305. SiO₂@NPEO has the smallest dissociation energy barrier, at 2.31023, which indicates that H₂O within the NPEO hydration layer can be dissociated more easily. A binding energy barrier exists for surfactants of the same type modifying nano-SiO₂, and it is higher than the dissociation energy barrier, indicating the presence of a hydration layer around the head group of the surfactant molecule. Consequently, the order of wettability strength of nanoparticles synergizing with different types of surfactants is SiO₂@SDBS₂ > SiO₂@DTAB > SiO₂@NPEO, consistent with the previous radial distribution curves results. Different surfactants have different wetting degrees on the coal-water interface. Meng et al. [52] found that the anionic surfactant SDBS solution significantly improved the hydrophilicity of the coal surface by atomic force microscopy (AFM). Xin et al. [53] carried out core flooding experiments and found that anionic surfactants reduced the number of nanoparticles at the water film interface and achieved the best enhanced oil recovery effect by repulsive balance.

Table 5. Binding energy barrier and dissociation energy barrier between SiO₂@surfactant and H₂O.

	E_{BARR}	E_{SSM}	E_{CM}	ΔE_1	ΔE_2
SiO₂@DTAB	1.5302	0.6484	-0.96034	0.8818	2.49054
SiO₂@SDBS	1.49311	0.60794	-1.38994	0.88517	2.88305
SiO₂@NPEO	1.54225	0.59089	-0.76798	0.95136	2.31023

3.2.3. Hydrogen bond numbers

The ability to form hydrogen bonding reflects wettability of the model system. The formation of hydrogen bonding is mainly controlled by electrostatic interactions. In the coal-water adsorption, the main hydrogen bond donor is hydrogen atoms in water molecules, and the main acceptor is the oxygen and nitrogen atoms in the oxygen-containing and nitrogen-containing groups in the coal. In order to better study the formation of hydrogen bonding, the intermolecular hydrogen acceptor distance less than 2.5 μm and donor-hydrogen-acceptor angle greater than 135° are used as geometric criteria for the determination of hydrogen bonding. The statistical results of simulated hydrogen bonding are shown in [Fig. 12](#).



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Fig. 12. Hydrogen bonds in each system.

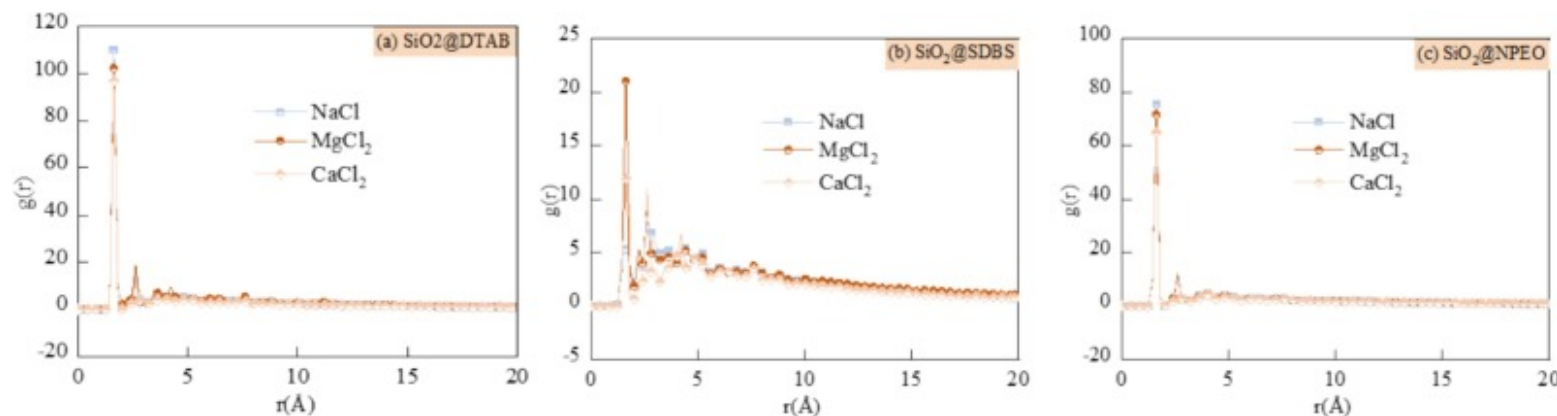
As shown in the figure, the addition of nanoparticles and surfactants in the system accelerates the formation of hydrogen bonding. The system with the largest number of hydrogen bonding is SiO₂@SDBS (1780 bonds, length 1.943 μm, and 152.294° angle). The anionic head group (sulfonic acid group) of SDBS has the strongest polarity, highest hydrogen bonding activity, and most significant synergistic effect with SiO₂, so the number of hydrogen bonding is the largest. The SiO₂@DTAB system is second. The DTAB cationic head group enhances the aggregation of water molecules through electrostatic interaction, resulting in more bonds than the SiO₂@NPEO system. The electronegativity of oxygen atoms in the NPEO polar head group (ether bonds) is lower than that of the key atoms in the SDBS and DTAB polar head groups, the polarity of the polyoxyethylene chain is weak, and the strength and number of hydrogen bonding with water, nano-SiO₂, and coal are limited. This is consistent with the wettability relationship obtained in [Section 3.2.2](#).

3.3. Wettability analysis of salt solution on synergistic nanoparticle surfactants

Inorganic salts in water can be ionized into metal cations and non-metal anions like chloride ions, which play a role in charge shielding for surfactants-modified nanoparticles with opposite charges in solution, thereby exerting a significant influence on the adsorption behavior of nanoparticle synergistic surfactants. The effects of Cl^- , Na^+ , Mg^{2+} and Ca^{2+} on the hydration of different surfactants-modified nanoparticles were analyzed mainly using the radial distribution function $g(r)$.

3.3.1. Effect of non-metallic anions on the hydration of synergistic nanoparticles surfactants

Fig. 13 presents the radial distribution curves between SiO_2 @surfactants and Cl^- in NaCl, MgCl_2 , and CaCl_2 solutions. The polar head group of the cationic surfactant DTAB is positively charged and readily engages in electrostatic interaction with the nonmetallic anion Cl^- . As a result, its peak $g(r)$ with Cl^- is the highest, indicating the strongest interaction. SDBS, an anionic surfactant, exhibits electrostatic repulsion with Cl^- . NPEO is a nonionic surfactant, which does not have direct electrostatic interaction with Cl^- . Nevertheless, its polyoxyethylene chain forms hydrogen bonding with water, thereby altering the microstructure and ion distribution within the solution. Fig. 13(a) shows that the peak of interaction between SiO_2 @DTAB and Cl^- in the NaCl solution is the lowest, indicating weak interaction. This is mainly because NaCl contains half as many Cl^- ions as MgCl_2 and CaCl_2 , which leads to a stronger electrostatic shielding effect in MgCl_2 and CaCl_2 on DTAB head groups. The interactions between SiO_2 @SDBS, SiO_2 @NPEO and Cl^- as shown in Fig. 13(b) and Fig. 13(c), are weaker than SiO_2 @DTAB shown in Fig. 13(a). SiO_2 @SDBS has the strongest interaction with Cl^- in MgCl_2 , while SiO_2 @NPEO exhibits the strongest interaction with Cl^- in CaCl_2 . The peak position of its radial distribution curve with Cl^- is more complicated in the latter case, which caused by the alteration of water molecule structure.



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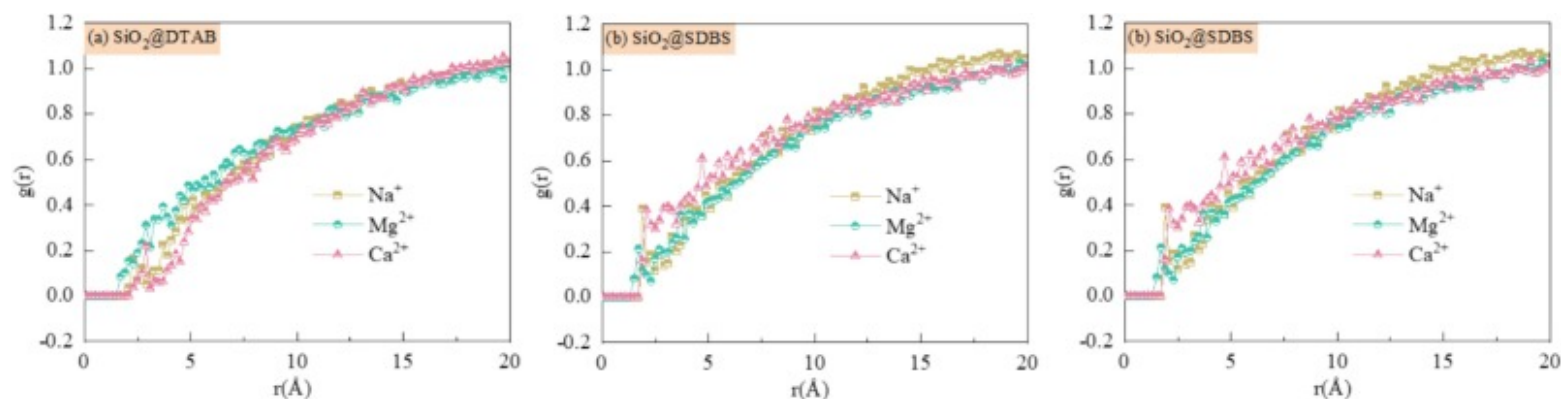
Fig. 13. Radial distribution curves between SiO_2 @surfactant and Cl^- .

The different cations present in NaCl, MgCl_2 , and CaCl_2 affect the interaction between SiO_2 @surfactants and Cl^- . Due to variations in ionic radius, charge densities, and other properties, different cations exhibit distinct binding affinities for Cl^- . This, in turn, affects the distribution of Cl^- within the NaCl, MgCl_2 , and CaCl_2 solutions, as well as the interaction with modified surfactants. The relatively high charge density of Ca^{2+} endows it with a strong binding capacity for Cl^- . This may cause Cl^- to distribute more tightly around the SiO_2 @surfactant, resulting in a higher peak of the radial distribution curves and closer contact with the surfactant, which accords with the observations in both Fig. 13(a) and Fig. 13(c). Whereas Na^+ has a larger ionic radius and weaker binding ability, the distribution of Cl^- is relatively more dispersed, resulting in a lower radial distribution curve peak. However, Fig. 13(b) is an exception, mainly because according to the Hofmeister sequence, different ions have distinct effects on the micellization and aggregation of surfactants. For SiO_2 @SDBS, it is likely that the relatively strong salting-out capacity of Mg^{2+} in the Hofmeister sequence plays a role, which makes the micellar structure of SiO_2 @SDBS more compact, thereby affecting the distribution of Cl^- around the micelles, and altering its radial distribution curve correspondingly.

3.3.2. Effect of metal cations on the hydration of synergistic nanoparticle surfactants

Metal cations primarily affect the head group of the anionic surfactant SDBS. Fig. 14 presents the radial distribution curves between SiO_2 @surfactant and metal ions. Fig. 15 depicts the radial distribution curves between SiO_2 @surfactant and inorganic

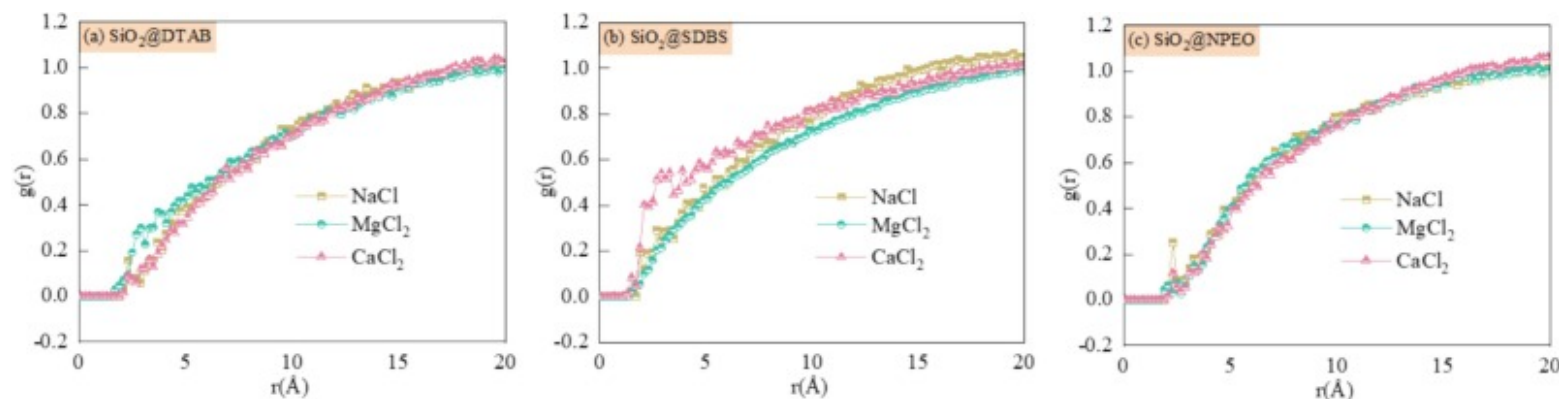
salt solution. In both figures, the values of $g(r)$ are all gradually increase. In the case of $\text{SiO}_2\text{@DTAB}$, DTAB is a cationic surfactant, which imparts a positive charge to the surface of nano- SiO_2 . For Na^+ in NaCl solution, it has electrostatic repulsion. However, for Mg^{2+} and Ca^{2+} in MgCl_2 and CaCl_2 solutions, electrostatic attraction occurs. When the surfactant coverage on the nano- SiO_2 surface changes, it breaks the balance of electrostatic interactions. Due to the electrostatic attraction exerted by the metal ions, those ions approach the surface of the $\text{SiO}_2\text{@DTAB}$, increasing $g(r)$. However, being an anionic surfactant, they impart a negative charge to the surface of nano- SiO_2 . They have weak electrostatic attraction with Na^+ , and strong electrostatic attraction to Mg^{2+} and Ca^{2+} . With the increase of metal ions in the system, driven by electrostatic attraction, these ions tend to aggregate to a greater extent on the surface of $\text{SiO}_2\text{@SDBS}$, which increases $g(r)$. For $\text{SiO}_2\text{@NPEO}$, the polyoxymethylene chains of NPEO form complexes with the metal ions. Ca^{2+} and Mg^{2+} are more prone to forming stable complexes with the polyoxymethylene chains than Na^+ is. As the complexation reaction progresses, the local concentration of metal ions in the vicinity of $\text{SiO}_2\text{@NPEO}$ rises, thereby increasing $g(r)$.



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Fig. 14. Radial distribution curves between $\text{SiO}_2\text{@surfactant}$ and metal ions.



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Fig. 15. Radial distribution curves between SiO₂@surfactant and inorganic salt.

Metal cations alter the hydration of SiO₂@surfactant. The aggregation of Na⁺ attracts more water molecules, thereby thickening the hydrated layer. However, this process also changes the structure of the hydrated layer, because Na⁺ undergoes hydration with the water, competing with the surfactant. The charge density of Mg²⁺ and Ca²⁺ is higher than that of Na⁺ (charge density depends on charge number and ion radius), causing a stronger electrostatic effect on SiO₂@surfactant. Mg²⁺ interacts with the polar groups of the surfactant, changing the strength and number of hydrogen bonding between water and surfactant in the hydration layer, and altering the hydration environment surrounding the surfactant, affecting reconstruction of the hydration layer and diffusion of the water. The ionic radius of Ca²⁺ (about 0.099 nm) is larger than that of Mg²⁺ (about 0.065 nm), which leads to different steric hindrance effects around the surfactant. Ca²⁺ occupies a larger space, making the distribution of water molecules around the surfactant more uneven, which affects the uniformity and stability of the hydration layer. In inorganic salt solutions containing ions such as Cl⁻, Na⁺, Mg²⁺ and Ca²⁺, Ca²⁺ interacts with the other ions in either a synergistic or competitive manner. When a large quantity of Cl⁻ is present, the aggregation behavior of Ca²⁺ can be affected, which in turn alters the way it affects the hydration of the surfactant.

As the concentration of the inorganic salt solution increases, the ionic atmosphere within the solution is altered. At low concentrations, the SiO₂@surfactant exists in a relatively independent state. As the concentration increases, however, the surfactant aggregates due to the interaction between the ions and surfactant molecules, as well as inter-ionic interactions. This

aggregation modifies the microenvironment surrounding the surfactant, attracting inorganic salt ions, and thereby increasing $g(r)$. In addition, a high concentration of inorganic salt solution can alter the microstructure of the solution, for instance changing the arrangement of water molecules within the solution, as well as the distribution of hydrogen bonding. With SiO_2 @surfactant, the solution structure changes accordingly, which affects the distribution of the inorganic salt ions. At higher concentrations, they gradually shift from a relatively uniform distribution into a state of aggregation around the SiO_2 @surfactant, resulting in increased $g(r)$.

In summary, the radial distribution curves of different types of surfactants modified by non-metallic anions, metal cations, inorganic salts and nano- SiO_2 reveal the following key findings: For MgCl_2 , the interaction with Cl^- was the strongest in the SiO_2 @SDBS system. In the SiO_2 @DTAB and SiO_2 @NPEO systems, the metal cations can also interact strongly with the surfactant to change the system structure. For CaCl_2 , in the SiO_2 @NPEO system, the interaction with Cl^- is the strongest. At the same time, its metal cations can also significantly affect the structure of the system through strong electrostatic interaction or complexation in different surfactant systems. Therefore, compared with NaCl, MgCl_2 and CaCl_2 can significantly improve the coal wettability in the interaction with different types of surfactants modified by nano- SiO_2 . This is consistent with the conclusion that the effect of inorganic salt additive CaCl_2 on wetting coal obtained by surface tension, contact angle test and moisturizing experiment analysis by Liu et al. [54] and Wang et al. [55].

3.4. The dynamic mechanism of synergistic nanoparticle surfactants under the influence of aqueous/salt solutions

Water can alter the state and motion of nanoparticles and surfactant molecules through solvation, diffusion, and migration processes and interfacial behavior. Under solvation, the water forms a distinct solvated layer around the nanoparticles and surfactant molecules, which not only affects the dispersion of the nanoparticles, but also modifies their effective size and surface properties, thereby influencing their dynamic behavior. Regarding diffusion and migration, nanoparticles and surfactant molecules undergo diffusion via Brownian motion in aqueous solution, driven by the thermal motion of water molecules. Moreover, the presence of water modifies the interfacial tension, charge distribution, and other properties, which significantly affects the adsorption and desorption kinetics of the nanoparticles and surfactants at the interface.

In inorganic salt solution systems, multiple interactions exist between the solute molecules and water molecules, including electrostatic interactions, hydration processes, and their influence on the solvent structure. When inorganic salts are dissolved in water, dissociation occurs, producing solute ions which interact with the water in different ways, which substantially alters the

hydration properties of the solution. For instance, in a solution containing NaCl, Na^+ and Cl^- shield the surface charges of nanoparticles, reducing the distance between them, and accelerating their aggregation. As for surfactants, alterations in ionic strength affect their critical micelle concentration (CMC). In general, as the ionic strength increases, the CMC of the surfactant decreases, which is because the electrostatic interaction between the counter-ion and the head group of the surfactant ion reduces the repulsive force among the head groups, making it easier for the surfactant molecules to aggregate and form micelles. Some metal ions (such as Ca^{2+} , Na^+ and Mg^{2+}) play a bridging role between the nanoparticles and surfactants. These metal ions are capable of complexing with both the functional groups present on the surface of the nanoparticles and the hydrophilic head groups on the surfactant, forming ionic bridges.

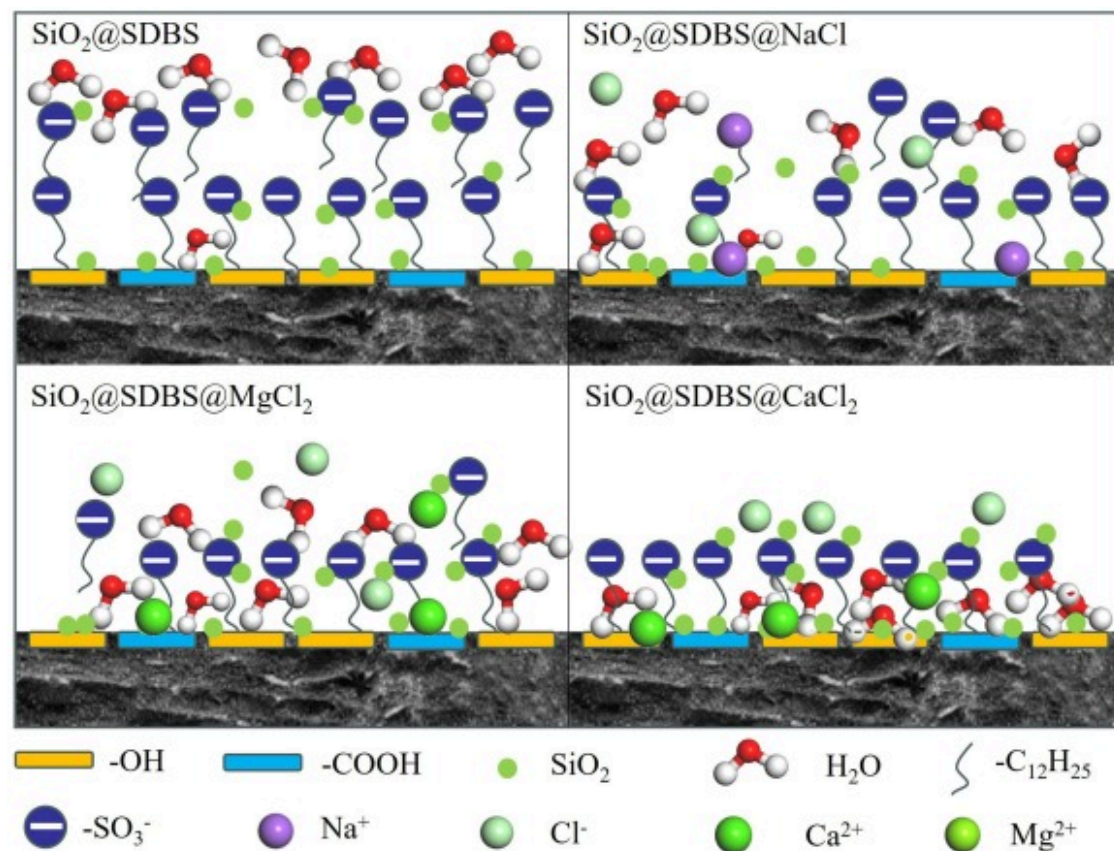
When nano- SiO_2 synergized with surfactant under the influence of aqueous/inorganic salts acts on the coal body, reactions occur between the metal ions and the nano- SiO_2 , surfactant and coal surface. The nano- SiO_2 surface typically contains hydroxyl groups ($-\text{OH}$), which react with the metal ions in CaCl_2 and MgCl_2 salt solutions through complexation. The hydration effect is weaker in NaCl salt solution, in contrast, which does not easily react with the hydroxyl groups on the nano- SiO_2 surface. Generally, no ionic reactions occur with the DTAB and NPEO surfactants. SDBS, an anionic surfactant with a structure of $\text{C}_6\text{H}_4^-\text{SO}_3^-\text{Na}^+$, already contains Na^+ , and thus does not have an ionic reaction with Na^+ , but it does have an ionic reaction with Ca^{2+} , and Mg^{2+} , producing a precipitate. Coal has a complex structure that encompasses aromatic and aliphatic hydrocarbons, along with carboxyl and phenolic hydroxyl groups (Ar), which can undergo ionic reaction with the metal ions to generate corresponding metal salts. [Table 6](#) shows the ionic reactions that occur in the solution system. In a salt solution environment with nanoparticles synergized with surfactants acting on the coal surface, the coal wettability increases significantly compared to the action of an aqueous solution alone. This is because the presence of inorganic salts in the salt solution alters the surface charge and double layer structure of the nanoparticles, and also affects the critical micelle concentration of ionic surfactants. The aggregation behavior of the nanoparticles modifies their adsorption mode and distribution on the coal surface. Moreover, the hydrophilic groups of the surfactant interact with the water molecules in the salt solution, which in turn enhances the coal wettability.

Table 6. Ion reactions in solution system.

	Ion reaction
Nano- SiO_2 and metal ions	$2\text{Si-OH} + \text{Ca}^{2+} \leftrightarrow (\text{Si-O})_2\text{Ca} + 2\text{H}^+$
	$2\text{Si-OH} + \text{Mg}^{2+} \leftrightarrow (\text{Si-O})_2\text{Mg} + 2\text{H}^+$

Ion reaction	
SDBS and metal ions	$2\text{C}_{12}\text{H}_{25}\text{-C}_6\text{H}_4\text{-SO}_3\text{-Na}^+ + \text{Ca}^{2+} \rightarrow (\text{C}_{12}\text{H}_{25}\text{-C}_6\text{H}_4\text{-SO}_3)_2\text{Ca}\downarrow + 2\text{Na}^+$ $2\text{C}_{12}\text{H}_{25}\text{-C}_6\text{H}_4\text{-SO}_3\text{-Na}^+ + \text{Mg}^{2+} \rightarrow (\text{C}_{12}\text{H}_{25}\text{-C}_6\text{H}_4\text{-SO}_3)_2\text{Mg}\downarrow + 2\text{Na}^+$
Coal and metal ions	$2\text{R-COOH} + \text{Ca}^{2+} \leftrightarrow (\text{R-COO})_2\text{Ca} + 2\text{H}^+$ $2\text{R-COOH} + \text{Mg}^{2+} \leftrightarrow (\text{R-COO})_2\text{Mg} + 2\text{H}^+$ $\text{R-COOH} + \text{Na}^+ \leftrightarrow \text{R-COONa} + \text{H}^+$ $2\text{Ar-OH} + \text{Ca}^{2+} \leftrightarrow (\text{Ar-O})_2\text{Ca} + 2\text{H}^+$ $2\text{Ar-OH} + \text{Mg}^{2+} \leftrightarrow (\text{Ar-O})_2\text{Mg} + 2\text{H}^+$ $\text{Ar-OH} + \text{Na}^+ \leftrightarrow \text{Ar-ONa} + \text{H}^+$

Fig. 16 shows the wetting mechanism of SiO₂@SDBS under the influence of aqueous/inorganic salt. The interaction between SiO₂@SDBS and coal is mainly manifested in the hydrophobic interaction and hydrogen bonding between SDBS and SiO₂. The hydrophobic chain -SO₃⁻ of SDBS is adsorbed on the hydrophobic region of the SiO₂ surface through the van der Waals force, and -SO₃⁻ bonds to the -OH on the SiO₂ surface through hydrogen bonding. The -OH, which is not covered by SDBS, forms hydrogen bonding with -OH and -COOH on the coal surface, and the exposed -SO₃⁻ (negative charge) of the SDBS generates electrostatic attraction with the local positive potential points (such as protonated amino or hydroxyl protons) on the coal surface. Na⁺ (monovalent cation) reduces the negative charge repulsion between SiO₂@SDBS and -SO₃⁻ through charge neutralization, and promotes the close arrangement of SDBS molecules on SiO₂ surface. Na⁺ adsorbs onto the negative electric sites on the coal surface (such as -COO⁻), reducing the negative electric repulsion between the coal and SiO₂@SDBS, and promoting attraction between the two. Ca²⁺/Mg²⁺ (high charge density) forms stable complexes (M²⁺...O⁻) with the -SO₃⁻ (O⁻) of SDBS, SiO₂ surface -OH, and -COO⁻ of coal surface to create “SiO₂@SDBS-M²⁺-coal” bridging structures (M= Ca²⁺, Mg²⁺).



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Fig. 16. Wettability mechanisms of $\text{SiO}_2@\text{SDBS}$ under the influence of aqueous/inorganic salt.

4. Conclusions

In this study, molecular dynamics was used to explore the synergistic effect of nano- SiO_2 with anionic (SDBS), cationic (DTAB), and non-ionic (NPEO) surfactants on the coal wettability in aqueous/salt solutions, revealing their microscopic mechanisms of action in a salt ion environment. The main conclusions are as follows:

(1) The synergistic effect of nano-SiO₂ with surfactants significantly alters the adsorption and diffusion behavior of water molecules on the coal wall surface, which is directly related to the optimization of coal wettability. H₂O adsorption on the coal wall surface is enhanced with increasing pressure, and the order of adsorption capacity is as follows:

SiO₂@SDBS>SiO₂@DTAB>without surfactant>SiO₂@NPEO. The adsorption heat of H₂O gradually increases increasing pressure, indicating enhanced adsorption stability. The combination of nanoparticles with different surfactants accelerates H₂O diffusion on the coal wall surface. The SDBS anionic surfactant promotes diffusion the fastest, while NPEO nonionic surfactant is the slowest.

(2) From the perspective of microstructure and energy, the radial distribution curves of nano-SiO₂ synergized with the three surfactants all showed peaks at 1.61 Å and 2.61 Å, indicating that a hydration layer forms around the surfactant and nano-SiO₂. An analysis integrating PMF and the binding and dissociation energy barriers determined that the wettability of synergistic nanoparticle surfactants follows the order SiO₂@SDBS>SiO₂@DTAB > SiO₂@NPEO. This result confirms that the synergistic effect of anionic surfactants and nano-SiO₂ is more advantageous for coal wettability.

(3) The peak of Cl⁻ interaction between SiO₂@DTAB and NaCl is lower than that of MgCl₂ or CaCl₂, indicating weak interaction between them. The g(r) values between the SiO₂@surfactants and metal ions as well as inorganic salts are all increase gradually. In the case of NaCl solution, the peak of water molecule distribution is the most prominent with SiO₂@SDBS. In the MgCl₂ solution, the peak of SiO₂@DTAB appears earlier. Additionally, SiO₂@NPEO exhibits the strongest interaction with Cl⁻ in CaCl₂ solution. MgCl₂ and CaCl₂ significantly improved coal wettability in the interaction with SiO₂@surfactants compared with NaCl.

(4) This study provides a theoretical basis for understanding coal wettability at the molecular level, and also provides a new method for the application of nano-SiO₂ with different types of surfactants as fracturing fluid in aqueous/salt solution environments. In the future, combined with field tests, the practical effect of SiO₂@surfactant as fracturing fluid can be explored, considering the complex conditions of coal seam water, providing a certain basis for the optimization of hydraulic fracturing technology.

CRedit authorship contribution statement

Xiaona Yang: Writing – original draft, Software, Methodology. **Peng Xiao:** Project administration, Funding acquisition, Conceptualization. **Ruizhe Wang:** Writing – review & editing, Resources, Data curation. **Biao Hu:** Writing – review & editing,

Visualization, Validation. **Xinyang Zhang**: Supervision, Investigation, Formal analysis. **Enyao Li**: Visualization, Validation, Resources. **Ruowan Su**: Writing – review & editing, Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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